

Boosting Lightweight CNNs Through Network Pruning and Knowledge Distillation for SAR Target Recognition

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Abstract—Deep convolutional neural networks (CNNs) have yielded unusually brilliant results in synthetic aperture radar (SAR) target recognition. However, overparameterization is a widely-recognized property of deep CNNs, and most previous works excessively pursued high accuracy but neglected the requirement of model deployment in radar systems, i.e., small computations and low memory cost. Therefore, further research on lightweight CNNs for SAR target recognition is necessary. In this article, we devise an effective CNN with channel-wise attention mechanism for SAR target recognition and then compress the network structure and recover lightweight network performance through network pruning and knowledge distillation, respectively. The attention values produced by the network are utilized to evaluate the importance of convolution kernels, and unimportant kernels are pruned. In addition, a novel bridge connection based knowledge distillation method is proposed. Instead of directly mimicking the hidden layer output or artificially designing a function to extract the knowledge in hidden layers, bridge connections are introduced to distill internal knowledge via teacher network. Experiments are conducted on the moving and stationary target acquisition and recognition benchmark dataset. The proposed network has excellent generalization performance and reaches an accuracy of 99.46% on the classification of ten-class targets without any data augmentation. Furthermore, through the network pruning and knowledge distillation algorithm, we cut down 90% parameters of the proposed CNN while maintaining model performance.

Index Terms—Convolutional neural network (CNN), knowledge distillation, model compression, network pruning, synthetic aperture radar (SAR) target recognition.

I. INTRODUCTION

SYNTHETIC aperture radar (SAR) has been increasingly used in military and civilian fields, among which SAR automatic target recognition (ATR) is a core research topic. Traditional SAR target classification algorithms, including template matching [1] and model-based methods [2], rely heavily on manually designed feature extractors and require a large amount

of expert prior knowledge. The rise of deep learning technology has brought new research directions for SAR ATR.

Deep convolutional neural networks (CNNs) can learn key features from raw data, and have been successful in many fields, such as face recognition [3], target detection [4], and image classification [33]–[36]. Therefore, in recent years, some scholars have also applied CNNs to SAR target recognition and achieved impressive results on the moving and stationary target acquisition and recognition (MSTAR) benchmark dataset. Chen and Wang [5] made a preliminary attempt to apply CNNs to SAR target recognition. A single-layer CNN was used to learn features automatically from SAR images and achieved 84.7% accuracy on the classification of ten-class targets. Ding *et al.* [6] utilized CNN with domain-specific data augmentation operations for SAR target recognition under challenging conditions. To enhance SAR target recognition performance, works in [9], [10], [44] focused on better feature representation. Lin *et al.* [9] used novel convolutional highway units to extract deep feature representations for classification. Zhang *et al.* [10] added convolutional block attention module [11] to CNNs, making networks pay attention to important areas of SAR images. The idea of capsule networks was also applied to help capture spatial relationships specialized in classification between different entities in SAR images. Another research direction is to alleviate overfitting caused by insufficient SAR data [7], [8], [45], [46]. To reduce the number of free parameters and alleviate overfitting, an all-convolutional network (A-ConvNet) with sparsely connected layers was presented in [7]. The fine-tune strategy was used in [8] to address the problem of misclassification and overfitting. Based on ResNet [35] with 18 layers, [45], [46] solved the problem through deep transfer learning and dropout layers.

These CNNs are achieving increasingly higher accuracy. However, advanced CNNs, accompanied by numerous parameters and high computational complexity, are difficult to store and run in resource-constrained radar systems. Although some researchers have considered lightweight network structure design [7], [43] and fast convolution operations [43], few researchers specifically further compressed their proposed network to obtain a slimmer CNN model that is more suitable for deployment.

Mainstream model compression technologies for CNNs consist of network pruning, low-rank approximation, network quantization, and knowledge distillation [12], [13]. The main idea of network pruning is to remove relatively unimportant weights

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or neurons in a network to balance performance and computational costs [14]. The motivation of low-rank approximation is that the convolution kernel, a four dimensional tensor, could be replaced by an approximate tensor with fewer dimensions [15]. Quantization technology aims at reducing the precision of network weights, e.g., using fewer bits to represent weights with less accuracy loss [16]. Based on the teacher-student strategy, knowledge distillation methods train a small student network under supervision information from strong pretrained teacher networks [17]. An emerging trend is to combine these different model compression methods to improve performance. Network compression applications have been applied to SAR community in recent years. Min *et al.* [39] compressed a deep CNN by combining knowledge distillation and weight quantization for SAR target recognition. Chen *et al.* [40] introduced a quantified squeezing method based on pruning, which can find the appropriate squeezing ratio quickly for sensitive CNNs. Furthermore, network pruning and knowledge distillation complement each other well. The former prunes a trained network to obtain a slimmer network but with worse performance. The latter is suitable for improving the performance of a small student network, but a norm to determine the structure of teacher and student networks has yet to be established. Their combination could not only compress network architecture but also maintain performance. Zhang *et al.* [41] and Chen *et al.* [42] followed the idea of combining network pruning with knowledge distillation. Zhang *et al.* [41] focused on training a lossless lightweight CNN for SAR target recognition. They combined a general pruning method [20] and a common knowledge distillation method [17] to achieve network compression. Chen *et al.* [42] trained a slim SAR ship detection network with only 2.8M model size. They used the network pruning algorithm in [22] to compress network and a novel knowledge distillation method was proposed to transfer the feature inter-relationship knowledge to smaller network. Although our compression approach is also based on pruning and knowledge distillation, the differences are summarized as follows. Our pruning strategy is a new pruning method presented to compress the proposed CNN (or CNNs with squeeze-and-excitation block). For example, the methods in [41], [42] all use existing general pruning methods to prune their networks, that is, they did not propose pruning methods for their own networks. On the contrary, our proposed pruning strategy is only applicable to our proposed network, CA-Net, or the network with the squeeze-and-excitation (SE) module. The reason is that our proposed pruning strategy prune the network based on the SE module embedded in CA-Net, which can better retain the key information in the network. Our knowledge distillation method focuses on transferring the implicit knowledge in hidden layers and uses teacher network to make the information loss as little as possible. For example, Zhang *et al.* [41] uses the existing original knowledge distillation method, which transfers the probability distribution of the top category learned by the teacher network. Chen *et al.* [42] extracts the hidden layer knowledge based on a manually designed function, which is called feature inter-relationship knowledge. However, our proposed knowledge distillation method focuses on the transfer of hidden layer knowledge, and uses teacher network instead of manually

designed functions to extract the key information in the hidden layer, which strives to minimize the information loss.

Considering the characteristics of SAR images, model size, and performance, we first devise a new effective CNN combined with the channel-wise attention mechanism (CA-Net), which achieves remarkable results with fewer parameters. Then, an attention-based network pruning strategy is presented to compress the architecture of CA-Net. Given that the output of the attention module reflects relative importance of filters in each convolutional layer, we prune unimportant convolutional channels on the basis of the attention values to obtain a more compact network. To compensate for the performance penalty caused by network pruning, a novel and effective bridge connection based knowledge distillation algorithm is proposed to enhance the performance of pruned networks. The trained CA-Net is regarded as the teacher network, whereas the pruned network is regarded as the student network. We aim to distill and transfer internal knowledge of the teacher network into the student network.

The main contributions of this article are summarized as follows.

- 1) An effective CNN called CA-Net is devised for SAR target recognition in consideration of the characteristics of SAR images, model size, and performance.
- 2) An attention-based network pruning strategy is presented to compress the proposed CNN, and it produces better results than the comparison method under the same compression rate.
- 3) A novel knowledge distillation algorithm is proposed to distill and transfer knowledge from base network to pruned network. Instead of directly mimicking hidden layer output or artificially designing a function to extract knowledge in hidden layers, bridge connections are established to distill internal knowledge via a teacher network because a teacher has learned how to extract useful information and does not lose key information.

The rest of this article is organized as follows: Section II reviews the related works on network pruning and knowledge distillation. Section III details the CA-Net and our proposed network pruning and knowledge distillation methods. Experiments and analyses are presented in Section IV. Finally, Section V concludes this article.

II. RELATED WORK

A. Network Pruning

Network pruning has been proven to be an effective model compression technique for deep neural networks. Specific rules are designed to determine the importance of model weights, then, unimportant weights are pruned, and important weights are retained to keep model accuracy. Network pruning methods can be divided into two categories: individual weight pruning and structured pruning [18].

Individual weight pruning, also called unstructured pruning, aims at pruning unimportant weights or connections to make weight matrices sparse [19]. However, a disadvantage that cannot be ignored is that this pruning strategy does not change the network architecture, and acceleration can only be achieved on

dedicated libraries and hardware. By contrast, structured pruning methods prune channels/kernels and even layers. Structured pruning can change the network structure and is more deployment friendly than unstructured pruning methods. Channel-wise pruning is mostly used for CNNs. The method in [20], in which channels are pruned based on the l_1 -norm of the corresponding filter weight, is the most representative. Hu *et al.* [21] leveraged the average percentage of zeros in outputs to evaluate filter importance. Liu *et al.* [22] pruned unimportant filters on the basis of scaling factor of the batch normalization layer. Structured pruning methods are economical in deployment, therefore we present an attention-based structured pruning method to slim the network.

B. Knowledge Distillation

Hinton *et al.* [17] pioneered knowledge distillation methods in deep neural networks. As the output probability distribution of a teacher network reflects similarities between categories, a student gains improved generalization and accuracy by learning from the probability distribution compared with learning only from labels. Many researchers have developed the above method for various fields and tasks, such as image segmentation [23], object detection [24], and natural language processing [25]. A major research direction is to transfer implicit knowledge in hidden layers, and this kind of methods is called feature-based knowledge distillation. Romero *et al.* [26] used hidden layer outputs of a teacher network as hints to improve the performance of a thinner and deeper student network. Zagoruyko and Komodakis [27] introduced the concept of attention transfer and converted activation values of hidden layers to an attention map. Yim *et al.* [28] used Gramian matrix to map the data flow process in a teacher network into a matrix, which contains knowledge about the flow of solution procedure. Xu *et al.* [29] conducted knowledge distillation based on the framework of conditional adversarial networks. Tung and Mori [30] presented pairwise activation similarities within each input mini-batch to supervise the training of a student network with a trained teacher network. Heo *et al.* [31] transferred activation boundaries from a point of whether a neuron is activated or not.

These methods provide good and interesting ideas but more or less drop and restrain some information. For example, the knowledge distillation method in [31] transfers feature activation as a sign according to a specific amplitude threshold; thus, the student cannot mimic concrete activation values. The knowledge distillation method in [27] compresses feature maps as an attention map by calculating the sum of squares of activation values across the channel dimension; thus, the full feature representation of images cannot be transferred to the student. Therefore, we propose an improved distillation method to transfer implicit knowledge in hidden layers while losing as little information as possible.

III. METHOD

In this section, the whole procedure of training and compressing CA-Net is explained in detail. The pipeline of our method is shown in Fig. 1. First, we introduce the framework of

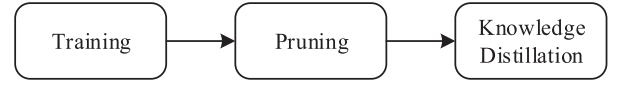


Fig. 1. Pipeline of our method.

TABLE I
FRAMEWORK OF CA-NET

Layer	Filter size	Filter number	Stride	Output size
Input	-	-	-	$1 \times 88 \times 88$
SE-Conv block 1	5×5	16	1	$16 \times 84 \times 84$
MaxPooling	2×2	-	2	$16 \times 42 \times 42$
SE-Conv block 2	5×5	32	1	$32 \times 38 \times 38$
MaxPooling	2×2	-	2	$32 \times 19 \times 19$
SE-Conv block 3	3×3	64	1	$64 \times 19 \times 19$
MaxPooling	2×2	-	2	$64 \times 9 \times 9$
SE-Conv block 4	3×3	128	1	$128 \times 9 \times 9$
MaxPooling	2×2	-	2	$128 \times 4 \times 4$
SE-Conv block 5	3×3	256	1	$256 \times 4 \times 4$
1×1 Convolution	1×1	10	1	$10 \times 4 \times 4$
Global AvgPooling	-	-	-	$10 \times 1 \times 1$
Softmax	-	-	-	$10 \times 1 \times 1$

CA-Net. Second, we describe how to prune CA-Net based on attention values to obtain a tiny network. Finally, we use a novel knowledge distillation method to improve the performance of the tiny network.

A. Framework of CA-Net

The framework of CA-Net is given in Table I (assuming that the input size is $1 \times 88 \times 88$). We consider following aspects to design CA-Net for SAR target recognition.

- 1) Large receptive fields can relieve the effects of speckle noise in SAR images to a certain extent. In CNNs, deeper layers have larger receptive fields. The receptive field of the first several layers is relatively small; thus, 5×5 convolution kernels are applied in the first two layers to increase receptive field considerably and reduce the size of feature maps rapidly. However, 5×5 convolution kernels introduce too many parameters; thus, 3×3 convolution kernels are utilized in the other layers.
- 2) We add the channel-wise attention mechanism, SE block [32], to the network. This module has been proven to improve model performance effectively and only introduces a few parameters. Specifically, the SE module can learn the importance of each feature channel, and important features are highlighted while suppressing useless features.
- 3) *Fully Connected* (FC) layers usually contain several parameters and easily result in overfitting. Therefore, FC layers are replaced by a 1×1 convolutional layer and a global average pooling (GAP) layer, which greatly alleviates model complexity. Furthermore, the GAP layer brings

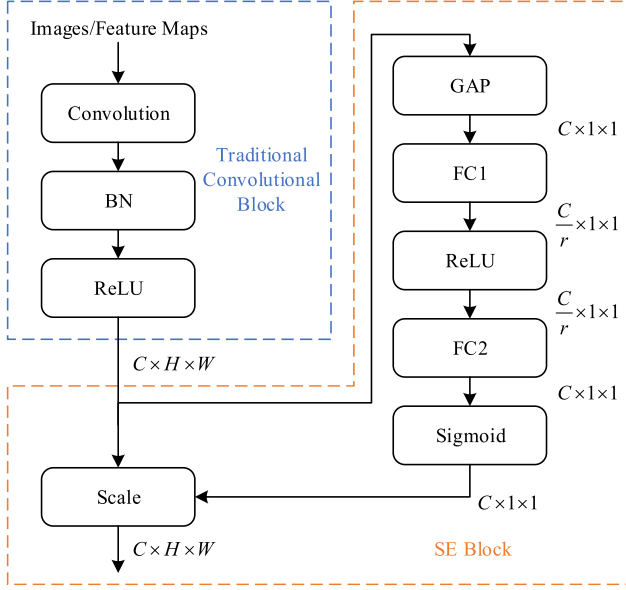


Fig. 2. SE-Conv block.

good robustness to the disturbance in images and allows the network to receive input images of different sizes.

CA-Net mainly consists of five SE-Conv blocks for feature extraction and 1×1 convolution + GAP for classification. A traditional convolutional block is composed of a convolutional layer, a batch norm (BN) layer, and a nonlinear activation function layer. The traditional convolutional block and SE block are combined as the SE-Conv block.

The architecture of the SE-Conv block is shown in Fig. 2. The traditional convolutional block produces feature maps $F \in \mathbb{R}^{C \times H \times W}$, where $H \times W$ represents the spatial size and C represents the number of feature channels. Then, the feature maps are squeezed across spatial dimensions to produce channel-wise feature descriptors through a GAP layer. The excitation operator consists of a dimensionality-reduction FC layer with a reduction ratio r , ReLU activation, a dimensionality-increasing FC layer, and sigmoid activation. The channel-wise feature descriptors are mapped to a set of channel attention values through an excitation operator. Finally, channel-wise multiplication is applied between the feature maps F and attention values to produce rescaled feature maps. The output shape of each layer is shown in Fig. 2.

B. Attention-based Structured Pruning Method

After training the CA-Net, the network pruning method is used to compress the architecture and obtain a slimmer network. The “parameter pruned ratio,” denoted by ppr , is defined as

$$ppr = 1 - \frac{\text{the number of pruned network's parameters}}{\text{the number of original network's parameters}}. \quad (1)$$

For example, $ppr = 0.9$ means that 90% original network's parameters are pruned.

Given that the attention values generated by SE module reflect relative importance of the filters in each convolutional layer, an attention-based channel-wise structured pruning strategy

Algorithm 1: Global Pruning Step.

Input: original network net_{ori} , target parameter pruned ratio ppr_{target} .

Output: pruned network.

1: Initialize parameters:

$net_{pruned} \leftarrow net_{ori}, ppr_{pruned} \leftarrow 0.$

2: **while** $ppr_{pruned} < ppr_{target}$ **do**

3: **for** $i = 1$ to 5 **do**

4: copy the network net_{pruned} and denote the copy as net_i ;

5: prune one filter with the smallest attention value in net_i 's i th conv-layer;

6: obtain the accuracy acc_i and parameter pruned ratio ppr_i of the pruned network net_i ;

7: **end for**

8: find the highest accuracy among $\{acc_1, acc_2, acc_3, acc_4, acc_5\}$ and record the corresponding index j ;

9: $net_{pruned} \leftarrow net_j, ppr_{pruned} \leftarrow ppr_j$;

10: **end while**

is naturally presented. The proposed attention-based pruning method is data driven; thus, all the training images must be fed into the trained CA-Net, and then the attention values could be obtained and saved.

The rule to determine which filters to prune within a single layer is provided. Let $v_{l,m}$ denote the attention value corresponding to the m th convolutional kernel in the l th convolutional layer. The filter importance in a single layer is estimated by the mean of $v_{l,m}$ over training images

$$\overline{v_{l,m}} = \frac{1}{N} \sum_{n=1}^N v_{l,m,n} \quad (2)$$

where N is the number of training images.

Then, we sort the filters in the l th convolutional layer by $\overline{v_{l,m}}$ and prune the filters with the smallest value. Fig. 3 shows an illustration of pruning channels based on channel-wise attention values. In fact, the weights in the BN layer and the SE block corresponding to the pruned channels must also be pruned, which are not drawn in Fig. 3 for convenience.

We have specified how to prune filters in a single layer in the above explanation, and then we introduce the global pruning strategy, which solves the problem of pruning filters across the whole network. Instead of evaluating layer's sensitivity or artificially set threshold, a greedy algorithm is presented to prune the whole CA-Net automatically and iteratively. In each iterative pruning turn, we generate five copies of the network to be pruned, and select different layers in each copy network to prune one filter with the smallest attention value. Then, we choose the pruned copy network that achieves the best performance as the pruning result in this pruning turn. The global pruning step of our method is shown in Algorithm 1. Finally, we can obtain the pruned networks, whose architecture is more compact.

The pruning method in [37] also uses the attention value from the SE block to prune networks. However, this method and ours

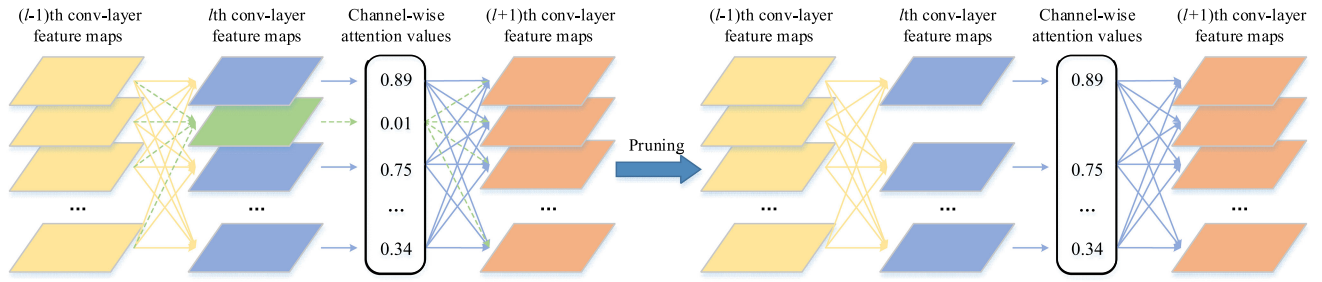


Fig. 3. Illustration of pruning channels based on channel-wise attention values within the l th conv-layer.

have two differences. One is that this method needs to add an SE block and remove it in each pruning iteration, thus requiring the pruned network to be fine-tuned in each iteration. In our method, only the weights in the SE block corresponding to pruned channels are pruned in each pruning iteration, and the pruned network does not need to be fine-tuned. Another difference is that given the compression ratio, the method in [37] just prunes the same percentage of channels in each layer, which is rough. On the contrary, our method prunes the network from a global perspective, which means that given just the compression ratio, our method can automatically select and prune the channels with minimal effect on performance, from the whole network. In other words, our method does not require artificially determining how many channels to prune in each layer.

C. Bridge Connection Based Knowledge Distillation

In network pruning, the pruned model is usually fine-tuned to restore performance. However, the effect of fine-tuning is limited due to the impaired feature extraction ability of the pruned model. A better choice is to transfer knowledge from the base network to the pruned network through knowledge distillation, which can compensate for the fitting ability of the pruned model to improve performance.

In our knowledge distillation method, the implicit knowledge of intermediate layers is extracted with the help of a pre-trained teacher network rather than an artificially defined function. Teacher networks have acquired the ability to extract useful information from the output of a hidden layer, inspiring us to establish bridge connections to maximize the ability.

To illustrate our method, we refer to some conceptions, such as “hint layers” and “guided layers” in [26]. Feature-based knowledge distillation methods try to select a hidden layer in the teacher network (called hint layer) and a hidden layer in the student network (called guided layer) and make the feature outputs of both layers similar. In other words, the guided layer is respected to predict the output of the hint layer. These methods, including ours, aim to distill knowledge in hidden layers and transfer it to the student. In general, the principle of selecting a pair guided/hint layer is that they are located similarly in their respective networks.

We define bridge connection as the connection that connects a guided layer and the layers higher than its corresponding hint layer, as shown in Fig. 4. Bridge connection acts as a bridge between the student and teacher, and introduces a bridge branch

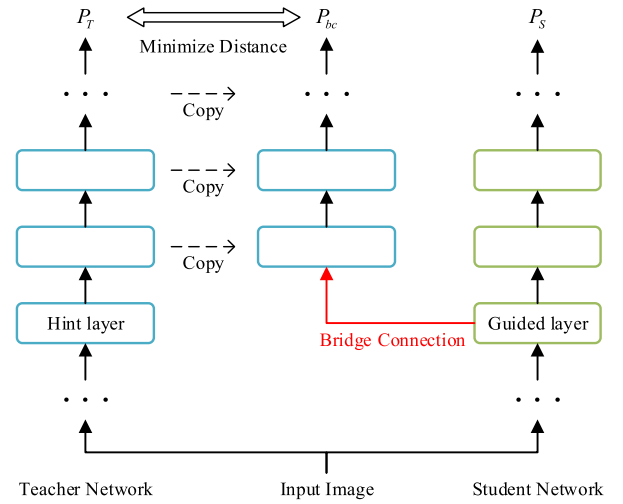


Fig. 4. Basic architecture of our knowledge distillation method. The teacher network is pre-trained and frozen.

that allows the output of the guided layer flow to the teacher. The bridge branch only uses the layers and weights of the pre-trained teacher and does not influence the feed-forward processes of the teacher network. The output of the guided layer is mapped to a class score through the bridge connection and teacher. In this manner, the problem of constraint similarity between two feature maps is translated into a simpler problem of constraint similarity between two class scores.

Although the knowledge distillation method in [38] also uses the “bridge” word, this method is different from ours in the definition and role of “bridge.” The bridge in [38] is more like a virtual connection between the guided layer and the hint layer, indicating that two ends of the bridge should be similar. This method uses MSE loss to reduce the feature representation differences between the guided layer and the hint layer. In our method, the “bridge” is more like a real pipeline between two networks, and feature maps from the student network are fed into the teacher through the pipeline. Moreover, the bridge connection plays a role in extracting key information from hidden feature maps.

Some mathematical descriptions and loss functions for our method are given in the following. A CNN is composed of some basic layers such as convolutional, pooling, and FC layers. Each basic layer can be represented as a function; thus, a neural network is actually a complex nested function. The teacher

network and the student network can be regarded as T and S , respectively.

When a bridge connection is established, both the teacher and student can be divided into two parts according to the hint/guided pair. For the teacher network, we denote the parts lower than the hint layer (including the hint layer) as T_L with parameters W_{TL} , and the part higher than the hint layer as T_H with parameters W_{TH} . T_L receives an image x as input and outputs feature maps, and then T_H takes the feature maps as input and outputs a class score P_T . Analogously, we can define S_L and S_H for student and obtain its output P_S

$$P_T = T(x) = T_H(T_L(x; W_{TL}); W_{TH}) \quad (3)$$

$$P_S = S(x) = S_H(S_L(x; W_{SL}); W_{SH}). \quad (4)$$

If the guided layer and its corresponding hint layer have different numbers of channels, 1×1 convolution must be added to the guided layer to match the channel numbers of the hint layer. Then the bridge connection can be established between the hint layer and guided layer. A bridge connection brings a bridge branch, and the branch output P_{bc} could be expressed as

$$P_{bc} = T_H(S_L(x; W_{SL}); W_{TH}) \quad (5)$$

where T_H is the tool to extract knowledge from feature maps and maps a big feature block to a probability vector with lower dimension. We can think of that during this mapping process, T_H helps dig and maximize the information in the feature block for it has been well trained.

Then, the feature difference between the pair hint/guided layer could be minimized by minimizing the distance between P_T and P_{bc} . Distance functions could be L1-norm, L2-norm, Kullback–Leibler (KL) divergence, or others. Under the inspiration of soft targets in [17], a “relaxation factor,” denoted by α , is introduced to soften both the outputs of the bridge branch and the teacher network, thus contributing to the optimization of the student network. The relaxation factor is used to soften the feature similarity constraint indirectly, whereas [17] uses temperature to soften the output probabilities of two networks. In this case, the knowledge distillation loss of one bridge branch is defined as

$$\begin{aligned} \mathcal{L}_{bc}(P_T, P_{bc}) &= \text{distance}(P_T^\alpha, P_{bc}^\alpha) \\ &= \text{distance}\left(\text{softmax}\left(\frac{z_T}{\alpha}\right), \text{softmax}\left(\frac{z_{bc}}{\alpha}\right)\right) \end{aligned} \quad (6)$$

where z_T and z_{bc} are the presoftmax outputs of the bridge branch and teacher, respectively.

When establishing multiple bridge connections, the output of the i th bridge branch is given by

$$P_{bc,i} = T_{H,i}(S_{L,i}(x; W_{SL,i}); W_{TH,i}). \quad (7)$$

The knowledge distillation loss with multiple bridge connections is computed as follows:

$$\mathcal{L}_{bc}(P_T, P_{bc}) = \sum_{i=1}^n \lambda_i \cdot \text{KL}(P_T^\alpha, P_{bc,i}^\alpha), i = 1, 2, \dots, n \quad (8)$$

where λ_i represents the weight of the i th bridge branch loss term.

Lastly, the total loss for training the student network is as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{H}(P_S, y_{\text{label}}) + \mathcal{L}_{bc}(P_T, P_{bc}) \quad (9)$$

where $\mathcal{H}$ denotes the cross entropy.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we first compare the performance of CA-Net with other CNNs. Second, the CA-Net is pruned on the basis of the attention-based structured pruning method. Finally, we show that “prune + knowledge distillation” can achieve higher accuracy than the traditional “prune + fine-tune” strategy. In addition, our proposed knowledge distillation method is compared with other methods to verify the effectiveness of the proposed method.

A. Experimental Data Description

Experiments are conducted on the MSTAR benchmark dataset, which is widely used for evaluating the performance of SAR target recognition algorithms. The MSTAR dataset is collected by a 10 GHz SAR sensor in a $0.3 \text{ m} \times 0.3 \text{ m}$ resolution spotlight mode and full aspect coverage. In accordance with the official advice, the samples at depression 17° and 15° are used for training and testing, respectively. The full dataset includes ten different categories of ground military vehicles: 2S1, BMP2, BRDM2, BTR70, BTR60, D7, T62, T72, ZIL131, and ZSU234. Some SAR images and the corresponding optical images of these ten types of targets are shown in Fig. 5. Referring to the existing literatures, the widely used experimental data settings include the ten-target MSTAR data without variants and the ten-target MSTAR data with variants. Compared with the ten-target MSTAR data without variants, the ten-target MSTAR data with variants is more complicated and more difficult to be recognized, the reason of which is that the variants used for test are not included in the training set. Details of these two experimental data setting are given in Tables II and III. All SAR images with 128×128 pixels are cropped at the center to 88×88 pixels. No image pre-processing algorithms and data augmentation operations are conducted. All experiments are conducted on a computer configured with Intel Xeon E5-2630 v4 CPU of 2.2 GHz, NVIDIA GeForce GTX 1080 Ti GPU, and 128 GB of memory on Ubuntu 18.04 Linux system, and the deep learning framework is PyTorch.

To train CA-Net, the following training settings are applied: SGD with an initial learning rate of 0.1, a weight decay of 0.0005, a momentum of 0.9, a minibatch size of 128, and a total number of epochs of 200. The learning rate is divided by 10 at 60, 120, and 160 epochs. The settings of fine-tuning and knowledge distillation are slightly different: initial learning rate 0.01; total epochs 100; learning rate divided by 10 at 50 and 80 epochs.



Fig. 5. Examples of SAR targets (left) and their corresponding optical images (right).

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES FOR TEN-TARGET EXPERIMENTAL
SETUP WITHOUT VARIANTS

Class	Train		Test	
	Depression	No. Images	Depression	No. Images
BMP2	17°	233	15°	196
BTR70	17°	233	15°	196
T72	17°	232	15°	196
BTR60	17°	255	15°	195
2S1	17°	299	15°	274
BRDM2	17°	298	15°	274
D7	17°	299	15°	274
T62	17°	299	15°	273
ZIL131	17°	299	15°	274
ZSU234	17°	299	15°	274

B. Network Performance Comparison

We start by evaluating the performance of CA-Net to verify its effectiveness and superiority. The reduction ratio r in the SE-Conv block is set to 4.

As given in Table IV, a confusion matrix is utilized to expose the performance of CA-Net on ten-target recognition. CA-Net achieves a remarkable result with 99.46% average accuracy rate. Even without any data augmentation, CA-Net does not suffer severe overfitting, demonstrating that CA-Net can extract pivotal features and has robust to the change of depression.

Then, we compare CA-Net with other CNNs, including two classic CNNs for optical image classification (VGG [34] and MobileNet [36]) and six CNNs designed for SAR target recognition (DeepSAR [8], A-ConvNet [7], AM-CNN [10], Capsule [44], ResNet-DTL [45], and ResNet-dropout-joint loss [46]), in terms of number of parameters, multiply-accumulate operations (MACC), and average classification accuracy. The experimental results are given in Table V. Among the compared CNNs, the accuracies of VGG and MobileNet are obtained by our experiments, whereas that of others is directly quoted from their papers.

Given that the data augmentation technique can boost classification accuracy, some authors only give the accuracy with data augmentation. Therefore, to evaluate the strength of various CNNs objectively, the symbol * in Table V is used to denote that the accuracy is obtained under data augmentation.

From Table V, we can clearly find that the classic CNNs, including VGG and MobileNet, cannot achieve higher accuracy than the CNNs devised for SAR target recognition. The main reason is that these networks have so many parameters that they get caught up in severe overfitting.

Among the networks devised for SAR target recognition, most of them can reach an accuracy of over 99%. However, the achievements of A-ConvNet, AM-CNN, and Capsule rely on data augmentation. Among the results without data augmentation, our proposed CA-Net obtains the second highest accuracy with 0.21% slightly lower than that of ResNet-dropout-joint loss. In terms of number of parameters and MACC, CA-Net is significantly smaller than majority of the compared CNNs. Moreover, compared with the proposed CA-Net, Capsule and ResNet-dropout-joint loss respectively contain 2.8 and 6.2 times larger parameters and 16.1 and 3.4 times greater MACC.

Lightweight network architecture and excellent feature extraction ability make CA-Net perform excellently. In summary, our proposed CA-Net makes a remarkable trade-off among accuracy, MACC, and number of parameters.

C. Network Pruning Comparison

After training CA-Net, the architecture of CA-Net will be compressed through structured pruning.

We compare our proposed attention-based pruning method with a generic weight-based pruning method [20] and a generic attention-based pruning method [37], the results of which are reported in Fig. 6. In addition, the number of pruned filters, parameters, and MACC of the pruned networks are presented in Table VI.

At the early stage of pruning, redundant filters that have minimal effect on model performance are pruned, and over-fitting is alleviated, which makes the pruned network even reach higher accuracy than the original network.

With the increase in parameter pruned ratio, the performance of the pruned model is gradually degraded at an accelerating rate. In the three network pruning methods, a rapid drop in accuracy occurs when the parameter pruned ratio reaches up to some certain critical value. The critical values of our method, the attention-based pruning method in [37], and the weight-based

TABLE III
NUMBER OF TRAINING AND TEST SAMPLES FOR TEN-TARGET EXPERIMENTAL SETUP WITH VARIANTS

Dataset	BMP2	BTR70	T72	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
Training (17°)	233 (C21)	233	232 (132)	255	299	298	299	299	299	299
Test (15°)	587 (C21, 9566, 9563)	196	582 (132, S7, 812)	195	274	274	274	273	274	274

TABLE IV
CONFUSION MATRIX FOR TEN-TARGET RECOGNITION USING CA-NET

Class	2S1	BMP2	BRDM2	BTR70	BTR60	D7	T62	T72	ZIL131	ZSU234	Acc(%)
2S1	272	0	1	0	0	0	1	0	0	0	99.27
BMP2	0	195	0	0	0	0	0	1	0	0	99.49
BRDM2	0	0	274	0	0	0	0	0	0	0	100
BTR70	0	0	0	196	0	0	0	0	0	0	100
BTR60	0	0	0	0	192	0	0	0	0	3	98.46
D7	0	0	0	0	0	270	0	0	0	4	98.54
T62	0	0	0	0	0	0	273	0	0	0	100
T72	0	0	0	0	1	0	0	195	0	0	99.49
ZIL131	0	0	0	0	0	0	0	0	272	2	99.27
ZSU234	0	0	0	0	0	0	0	0	0	274	100
Average accuracy											99.46

TABLE V
PERFORMANCE COMPARISON FOR VARIOUS NETWORKS

Networks	Parameter	MAcc	Accuracy(%)
VGG16 [34]	6.51×10^7	5.05×10^9	95.54
MobileNet [36]	3.22×10^6	1.86×10^8	92.46
DeepSAR [8]	4.16×10^6	1.18×10^8	98.15
A-ConvNet [7]	3.03×10^5	3.78×10^7	99.13*
AM-CNN [10]	2.56×10^6	3.48×10^9	99.35*
Capsule [44]	1.26×10^6	6.38×10^8	99.79*
ResNet-DTL [45]	1.12×10^7	5.68×10^8	99.46
ResNet-dropout-joint loss [46]	2.79×10^6	1.36×10^8	99.67
CA-Net	4.48×10^5	3.96×10^7	99.46

*Denotes that the accuracy is obtained with data augmentation.

TABLE VI
NETWORK PROPERTIES AT DIFFERENT PRUNED RATIOS

Parameter Pruned Ratio	Parameters	Number of Pruned Filters	MAcc
0	4.48×10^5	0	3.96×10^7
0.1	4.03×10^5	34	3.89×10^7
0.2	3.59×10^5	68	3.82×10^7
0.3	3.14×10^5	103	3.76×10^7
0.4	2.69×10^5	138	3.69×10^7
0.5	2.24×10^5	172	3.61×10^7
0.6	1.79×10^5	208	3.49×10^7
0.7	1.35×10^5	245	3.18×10^7
0.8	8.92×10^4	294	2.95×10^7
0.9	4.48×10^4	348	2.61×10^7
0.95	2.24×10^4	385	1.46×10^7

pruning method are 0.35, 0.3, and 0.55, respectively, but it cannot indicate which network pruning method is more superior. In general, fine-tuning stage follows the pruning stage to compensate for the performance penalty, and we use the accuracy of the fine-tuned network to evaluate the superiority of the network pruning method.

As shown in Fig. 6, fine-tuning the pruned network can effectively enhance the performance of the pruned network. For our proposed pruning method, the accuracy of the fine-tuned network is higher than 99% when the parameter pruned ratio is less than 0.7.

Furthermore, at the same parameter pruned ratio, our proposed attention-based pruning method produces higher accuracy

after fine-tuning than the other two comparative pruning methods, proving that the proposed pruning method is more suitable for pruning CA-Net. Although the pruning method in [37] also prunes networks based on the attention values generated by the SE block, it just prunes the same percentage of channels in each layer, which causes those important features, especially in the lower layers, is easily pruned during the early pruning. On the contrary, our method can automatically select and prune the channels with minimal effect on performance from the whole network under the given parameter pruned ratio. Fig. 7 shows the number of convolution channels under different parameter

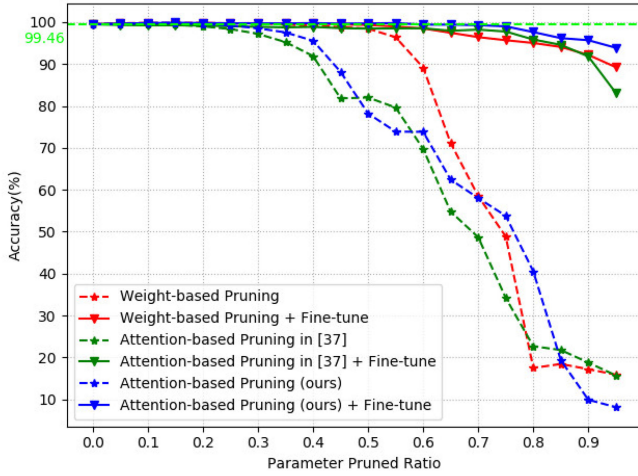


Fig. 6. Accuracy of pruned models under different parameter pruned ratios and different pruning methods.

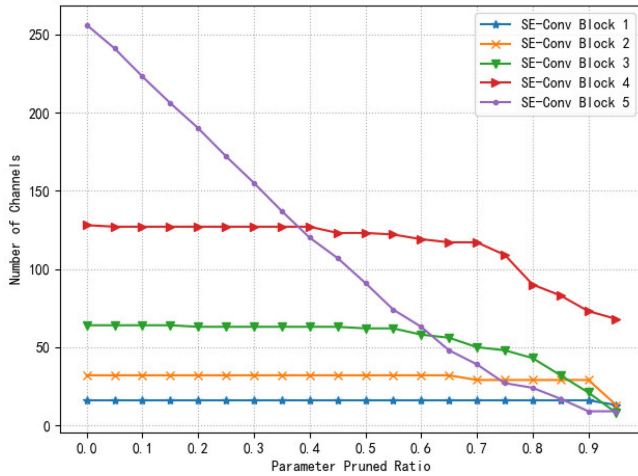


Fig. 7. Number of convolution channels under different parameter pruned ratio in the pruning method.

pruned ratio in our method. It can be seen that the channels in higher layers are pruned first, which means that these layers have more redundancy and have less impact on performance. When the parameter pruned ratio is larger, the channels in lower layers will start to be pruned, indicating that these layers have a greater impact on performance than the higher layers. This result also explains why our method achieves higher accuracy than the pruning method in [37].

The pruned network cannot recover to the accuracy of the original CA-Net by fine-tuning when the parameter pruned ratio is greater than 0.6. The main reason is that numerous convolutional filters are pruned, resulting in the decrease in the fitting ability of the network.

D. Knowledge Distillation Comparison

When the parameter pruned ratio is too large, the pruned network cannot recover to the original performance by fine-tuning. Therefore, the knowledge distillation technology is utilized to improve the performance of the pruned network further.

TABLE VII
KNOWLEDGE DISTILLATION RESULTS FOR VARIOUS DISTANCE FUNCTIONS AND RELAXATION FACTOR VALUES

α distance	1	2	3	4
L1	98.81	98.93	99.05	98.95
L2	98.55	98.74	99.13	98.71
KL	98.48	98.81	99.38	99.2

TABLE VIII
KNOWLEDGE DISTILLATION RESULTS FOR ESTABLISHING BRIDGE CONNECTIONS AT DIFFERENT LAYERS

Bridge connections	in which layers ?	λ_i	Accuracy(%)
Single	1	1	99.11
Single	2	1	98.87
Single	3	1	99.05
Single	4	1	98.96
Single	5	1	99.12
Multiple	1,5	1,1	99.15
Multiple	2,4	1,1	98.97
Multiple	1,3,5	1,1,1	99.18
Multiple	1,2,3,4,5	1,1,1,1,1	99.02

TABLE IX
KNOWLEDGE DISTILLATION RESULTS FOR VARIOUS WEIGHTS SETTINGS OF BRIDGE BRANCH LOSSES

Bridge connections	in which layers ?	λ_i	Accuracy(%)
Multiple	1,3,5	2,1,1	99.19
Multiple	1,3,5	1,2,1	98.93
Multiple	1,3,5	1,1,2	99.24
Multiple	1,3,5	2,1,2	99.3
Multiple	1,3,5	4,1,4	99.38
Multiple	1,3,5	6,1,6	99.21

In our knowledge distillation experimental setup, the trained CA-Net is regarded as the teacher network, and the pruned network is regarded as the student network. Several factors influence the distillation results. Thus, some experiments with different configurations are conducted to reveal the specific effects of each factor and the effectiveness of the proposed method. The pruned CA-Net with a parameter pruned ratio of 0.9 is used as the student network in these experiments.

1) *Distance Function and Relaxation Factor α* : In this experiment, we explore which distance function is the best in our knowledge distillation method and seek the most appropriate relaxation factor value α .

We use the accuracy of fine-tuning the pruned network obtained using our proposed attention-based pruning method with a parameter pruned ratio of 0.9 as the baseline accuracy, which is 95.67%. As given in Table VII, all the results are higher than the baseline accuracy, which shows that the results of knowledge distillation are higher than the results of fine-tuning even under

TABLE X
RECOGNITION ACCURACY COMPARISON FOR VARIOUS KNOWLEDGE DISTILLATION METHODS AND VARIOUS PARAMETER PRUNED RATIOS
USING THE TEN-TARGET DATA WITHOUT VARIANTS

Parameter Pruned Ratio	Pruned Network	Prune + Fine-tune	Prune + Knowledge Distillation						
			Soft Targets [17]	FitNet [26]	Attention Transfer [27]	Similarity Preserving [30]	Activation Boundary [31]	BridgeNet [38]	Ours
0	99.46	-	-	-	-	-	-	-	-
0.6	73.83	99.38	99.41	99.43	99.39	99.53	99.44	99.49	99.63
0.7	58.04	99.26	99.23	99.35	99.37	99.43	99.38	99.36	99.56
0.8	40.56	97.61	99.25	99.13	99.27	99.3	99.46	99.4	99.48
0.85	19.37	96.17	99.01	99.18	99.1	99.35	99.29	99.19	99.51
0.9	9.93	95.67	98.19	98.52	98.44	98.26	98.59	98.46	99.38
0.95	8.16	93.86	96.49	96.98	97.42	97.8	97.68	96.95	98.97

different hyperparameter configurations. Furthermore, when we do not relax both extra branch output and teacher output ($\alpha = 1$), L1-norm reaches higher accuracy. The best result is obtained when the distance function is KL, and the relaxation factor value α is 3.

2) *Establishing Bridge Connections in Which Layers*: Table VIII gives the knowledge distillation results of the single bridge and multiple bridge connections at different layers. The second row in Table VIII shows that bridge connections are established in which layers. For example, “1, 3, 5” denotes that three bridge connections are established between the student network and the teacher network at the last layers of SE-Conv block 1, SE-Conv block 3, and SE-Conv block 5, respectively. To facilitate comparison, the weights of the bridge branch losses λ_i are all set to 1.

When conducting single-layer knowledge distillation, layer 1 and layer 5 outperform the others. As for multilayer knowledge distillation, “1, 3, 5” achieves the best performance in all comparisons. Establishing bridge connections at all layers does not work very well, and its accuracy is lower than the best result of single bridge connection. The reason may be that distillation regulation is so strong that the student has little freedom to adjust.

3) *Weights of Bridge Branch Losses λ_i* : Following the best configurations above, we conduct experiments to reveal the effect of weight λ_i on distillation performance. As given in Table IX, bridge connections are frozen, and different combinations of λ_i are designed.

An interesting observation is that increasing the loss weights of both layer 1 and layer 5 is favorable for distillation performance, which is similar to the results in Table IX. However, if the weights are too large, such as “6, 1, 6”, we can see a performance reduction. Thus, the weights of bridge branch losses are set as $\lambda_1 = 4$, $\lambda_2 = 1$, and $\lambda_3 = 4$.

Our proposed knowledge distillation method is compared with other previous knowledge distillation methods, such as Soft Targets [17], FitNet [26], Attention Transfer [27], Similarity Preserving [30], Activation Boundary [31], and BridgeNet [38]. The training settings of all the knowledge distillation methods are the same. The weights of the teacher network are frozen, and the architecture of the teacher network has never been changed.

We compare the recognition accuracy for various knowledge distillation methods and various parameter pruned ratios using the ten-target data without variants and the ten-target data with variants, respectively. The results are given in Tables X and XI. All knowledge distillation methods can help the pruned networks achieve higher accuracy than that of the fine-tuned networks. It can be observed that the student even achieves higher accuracy than the teacher when the parameter pruned ratio is between 0.6 and 0.85. We think one reason is that the proposed knowledge distillation method can make the pruned network simulate the hidden layer features of the original network to enhance its feature expression ability, thereby improving its performance, that is, the proposed knowledge distillation method compensates for the feature extraction ability of the pruned network. Another reason is that the smaller pruned network shows better generalization performance on limited data. It should be noticed that this phenomenon is not a special case only in our work, the student network in some existing works can also perform better than the teacher network, such as the work in [41]. In addition, among all knowledge distillation methods, the proposed method reaches the best results at the same parameter pruned ratio, and the benefit of our method is particularly significant when the parameter pruned ratio is very large. Compared with our method, the accuracy of BridgeNet [38] becomes worse when the parameter pruned ratio is large. The reason is that BridgeNet only uses MSE loss on the output of the last hidden layer (SE-Conv block 5) in the teacher network and student network for feature constraints, which cannot guide the student network to learn lower layer features from the teacher. The results prove that the proposed knowledge distillation method can effectively extract key information from hidden feature maps and transfer knowledge from the teacher network to the student network.

As given in Tables X and XI, although the number of parameters is only 10% of the original CA-Net, the pruned network can also reaches an accuracy of 99.38% on the ten-target data without variants and 98.13% on the ten-target data with variants, which are close to the accuracy of the original CA-Net. All the experimental results show that our proposed knowledge distillation method is effective.

Finally, we successfully obtain a tiny CA-Net with less memory and computational cost and high accuracy. Furthermore,

TABLE XI
RECOGNITION ACCURACY COMPARISON FOR VARIOUS KNOWLEDGE DISTILLATION METHODS AND VARIOUS PARAMETER PRUNED RATIOS USING THE TEN-TARGET DATA WITH VARIANTS

Parameter Pruned Ratio	Pruned Network	Prune + Fine-tune	Prune + Knowledge Distillation						
			Soft Targets [17]	FitNet [26]	Attention Transfer [27]	Similarity Preserving [30]	Activation Boundary [31]	BridgeNet [38]	Ours
0	98.38	-	-	-	-	-	-	-	-
0.6	78.395	98.601	98.35	98.48	98.44	98.63	98.5	98.49	98.6
0.7	54.48	98.11	97.97	98.51	98.4	98.38	98.34	98.43	98.56
0.8	42.43	96.57	97.85	98.1	98.15	98.38	98.43	98.22	98.48
0.85	15.33	95.19	97.26	97.71	97.88	97.96	97.89	97.81	98.31
0.9	19.54	94.21	97.01	97.28	97.22	97.41	97.15	96.98	98.13
0.95	18.23	91.76	95.67	94.2	94.34	95.02	94.98	94.33	96.84

TABLE XII
PERFORMANCE COMPARISON FOR VARIOUS COMPRESSED MODELS

Model	Parameters	MACC	Accuracy(%)
Slim [40]	6.72×10^5	-	92
MCNN [39]	6.2×10^4	-	98.2
Compressed A-ConvNets [41]	3.7×10^4	7.77×10^6	99.13
CA-Net-0.9	4.48×10^4	2.61×10^7	99.38
CA-Net-0.95	2.24×10^4	1.46×10^7	98.97

CA-Net-0.9 (CA-Net with a parameter pruned ratio of 0.9) and CA-Net-0.95 are compared with some existing compressed CNNs for SAR target recognition, including Slim [40], MCNN [39], and Compressed A-ConvNets [41]. The results are given in Table XII. CA-Net-0.9 achieves the highest accuracy; meanwhile, the number of parameters is also smaller than Slim and MCNN. CA-Net-0.95 contains fewer parameters and obtains slightly lower accuracy than the compressed A-ConvNets. These results indicate that our method is superior to the comparison models to a certain extent.

V. CONCLUSION

Considering that most CNNs for SAR target recognition have too many parameters and high computational cost and are unsuitable for deployment on resource-constrained devices, such as radar systems, this article aims to train a lightweight CNN with small computation, low memory cost, and high accuracy. First, an effective CNN with channel-wise attention mechanism is devised for SAR target recognition. This network achieves remarkable accuracy without any data augmentation. Then, relatively unimportant convolutional filters are pruned to compress the network structure through an attention-based network pruning method, which is presented to compress the proposed CA-Net (or CNNs with squeeze-and-excitation block). However, the pruned network suffers from performance penalty. Therefore, we recover the performance of the pruned network through a novel bridge connection based knowledge distillation method. The proposed distillation method provides a general knowledge distillation framework for image classification and

plays a role in distilling the knowledge of the hidden layers with the help of the trained CA-Net and in transferring knowledge into the pruned network. The experiments conducted on the MSTAR dataset prove the effectiveness and superiority of all our methods. Lastly, we obtain a tiny CA-Net with a small number of parameters and high accuracy. We believe that the work in this article is conducive to promoting the deployment of CNNs in radar systems.

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